Electrification, telecommunications, and the finance-growth nexus: evidence from firm-level data

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Abstract

We explore variations in access to energy and telecommunications improvements across firms in 57 and 112 developing countries, respectively, between 2002 and 2014, and find that national investments in technological infrastructures provide a channel through which financial development promotes firm-level adoption of tangible innovations. In particular, when countries make large investments in their energy (electricity and natural gas) and telecommunications infrastructures, firms with greater access to financial resources are better positioned to benefit from these investments, experiencing significantly fewer power interruptions, more adoptions of power generators, and greater website use, all of which are related to higher firm-level growth in sales and employment.

Keywords: Energy-finance-growth nexus, electrification, IT, infrastructure, financial access, power interruption

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1. Introduction

A large empirical literature relates financial development to growth in economic activity (see King and Levine (1993), Rousseau and Wachtel (1998), and Levine et al. (2000), among many others),[1] with a number of recent papers linking financial development directly to energy consumption (Durusu-Ciftci et al., 2020; Wang and Gong, 2020). Another literature examines the introduction of general purpose technologies (GPTs) as fundamental shocks that can assist economies in transitioning to higher equilibrium growth paths (David, 1991; Bresnahan and Trajtenberg, 1995; Jovanovic and Rousseau, 2005). For the most part, however, these literatures live separate and parallel lives. We join these strands by asking if and how financial development can promote the adoption and effective use of GPTs, and in particular energy (electricity and natural gas) and telecommunications technologies. Given the relatively slow diffusion of electricity on a global level since its introduction 140 years ago and the enormous effects on everyday living for those enjoying its benefits, confirming this channel offers a new motivation for promoting financial development, and has important implications for encouraging the rapid diffusion of more recent technologies such as telecommunications.

To these ends, we use data from the *World Bank Enterprise Survey* (WBES) to explore the impact of national spending on energy and telecommunications infrastructures on firm-level constraints to investment, and how this impact varies with the extent of access to external finance. Specifically, we find firms with greater access to financial resources better positioned to mitigate the negative effects of power disturbances, including through the purchase of generators to complement an expanding but imperfect power grid, and to implement telecommunications technologies. We also find that firms with these capabilities coupled with greater access to finance experience higher growth in sales and employment. Our results suggest that firm-level financial access can play a key role in mediating the effectiveness of investments in energy

[1] Examples are Demetriades and Hussein (1996), Rousseau and Wachtel (1998), Beck (2002), Ang and McKibbin (2007), Baltagi et al. (2009), and Rousseau and D'Onofrio (2013). Levine (1997), See Levine et al. (2000), and Levine (2005) for useful surveys.

infrastructure, highlighting an important factor to consider when evaluating the potential impact of energy investments on economic outcomes. Our focus on the effective use of energy and telecommunication technologies reveals an additional conduit for finance that remains important today, and especially for developing economies.

The mechanism we study places financial access as central to productive activity through both the traditional "total factor productivity (TFP)" and "factor accumulation" channels (Pagano, 1993). More precisely, TFP is expected to rise as financial intermediaries fund more efficient projects, and typically in their entirety, but we elucidate an additional onduit which financial access enables firms to make complementary investments that render other public goods investments, such as electricity or telecommunications infrastructure, more effective. Moreover, our mechanism implies that general increases in financial development, through the mobilization of savings and the related spread of banks and other financial intermediaries across a country's populated areas, can only enhance the access of productive firms to the financing needed for complementary local investments such as power generators or computers and internet. The evidence we present thus has implications for multiple facets of the now-classic finance-growth nexus as it relates to energy.

Earlier work uses the WBES to examine the role of business constraints on firm activity within developing countries, and particularly for examining the importance of financial constraints in restricting growth within small and medium-sized firms (Beck et al., 2005; Ayyagari et al., 2008). There are very few studies, however, that examine factors that alter constraints faced by firms.[2] Here we exploit the time structure of the WBES to examine how a specific set of treatments, namely national spending on energy and telecommunications infrastructure, interacts with financing constraints in the short run to promote firm-level technological adoptions and improvements.

More generally, our findings contribute to a expanding literature that aims to identify various channels through which financial development can affect growth,

[2]One such study is Datta (2012), which uses changes in WBES responses to evaluate the impact of infrastructure spending on highways in India.

including factors such as trade openness (Beck, 2002; Alfaro et al., 2004), changes in firm dynamics (Arellano et al., 2012), and improvements in institutional quality (Rajan and Zingales, 2003; Chinn and Ito, 2006). We also complement studies that examine the effects of electrical outages (Allcott et al., 2016; Andersen and Dalgaard, 2013; Fisher-Vanden et al., 2015), expanded access to electricity and telecommunications technologies (Jensen, 2007; Dinkelman, 2011; Rud, 2012), improvements in infrastructure (Bougheas et al., 2000), and investment in private generators on industrialization, productivity, and firm-level business decisions (Steinbuks and Foster, 2010; Alby et al., 2013). One particular contribution of our paper is the combination of firm-level data with a substantial cross-country analysis. While most previous work in this literature involves either aggregate cross-country data or micro-level data from a particular country or small group of countries, our study demonstrates the importance of financial access at the level of the individual firm by utilizing data from a large number of developing countries across the globe.

The remainder of the paper is organized as follows. In Section 2 we present our main hypotheses and describe the data used to test them. Section 3 includes our strategy for identifying links between finance and infrastructure expansion, and shows that increased access to external finance at the firm level enhances the efficiency of country-level spending on electrical and telecommunications infrastructure by promoting early adoption of related technologies. In Section 4 we explore the link between firm-level adoption of these technologies and growth in sales and employment. We summarize and conclude in Section 5.

2. Motivation and data

Our question of interest can be broken down into two separate hypotheses. The first is that financial access is a key component to the successful and timely adoption of electrification and telecommunications technologies. The second is that adoption of these technologies facilitates firm-level growth. We propose a mechanism through which infrastructure investments made at the national level affect firms asymmetrically, with firms that enjoy greater access to external finance making better use of a given infrastructure investment. For example, improvements in internet access will be most useful to firms that can finance complementary improvements such as building and maintaining a website. We then test whether these improvements affect growth at the firm level across a broad panel of countries and time periods.[3]

We use data from two main sources. The first is the World Bank's *World Development Indicators* (WDI) database (2002-2014), which includes the value of country-level aggregate investments with a private participation component in various categories of infrastructure by year. The second is the *World Bank Enterprise Survey*, which includes survey data on factors that may influence firm-level production (Table A1 in the appendix lists the survey questions that we use). The two datasets allow us to assess the confluence of finance, electrification, telecommunications, and economic activity not only at the macro level (as is common in the finance-growth literature), but at the micro level as well. The WBES was first administered in 2002 and we use these data through 2014.[4]

The core of our empirical strategy is to examine within-country changes in infrastructure spending and how they affect outcomes for individual firms. To capture this within-country variation, we focus only on countries that were surveyed multiple times by the WBES. The annual data from WDI on infrastructure spending for individual countries is limited to cumulative spending on projects with private participation that were completed in that year. For example, our energy investment variable captures "infrastructure projects in energy (electricity and natural gas transmission and distribution) that have reached financial closure and directly or indirectly serve the public." The data only capture projects with some degree of private participation, but this still includes a significant portion of the total investment in infrastructure projects. Because of the cumulative nature of these spending variables,

[3]Dinkelman (2011) and Rud (2012) show that electrification impacts employment and industrialization in South Africa and India, respectively, but do not address the role of financial access in this process.

[4]The foundation for the World Bank Enterprise Survey can be traced back to a survey (the World Business Environment Survey) conducted in 1999 and 2000 by the World Bank. Since this survey is much less detailed than the WBES in its current format and exhibits significant discrepancies with the WBES, we focus on the WBES from 2002 to 2014.

we can expect to see an effect in the same calendar year for firms that can adopt new technologies quickly, although firms that are slower to adopt would not show rapid improvements. Our analysis is limited to identifying such short-run effects because the WBES does not provide longitudinal links across firms in different survey years.

[Table 1 here]

Table 1 reports descriptive statistics for two sub-samples of the WBES data. The classification of the data into sub-samples is necessary because the availability of country-level observations differs for investments in energy and telecommunications infrastructures, thereby limiting the number of firms that can be included in our empirical analysis for each investment type (Table A2 in the Appendix reports the geographic distribution of firm-year observations by size in the two sub-samples and Table A3 reports the number of firm-year observations for each country). As a result, we identify the two separate samples as the "energy sample" and the "telecommunications sample," which include firm-level data from 57 and 112 developing countries respectively. The "number of outages" variable measures the average number of power outages a firm experiences each month in a given survey period, and "average duration of outages" is the average length of an outage in hours. Outage hours are the product of these two measures. "Website use" is coded as unity if a company uses a website for business operations and zero otherwise. "Financial access" is coded on a scale of 0-4, with 4 representing the highest degree of financial access, and comes from a firm's self-reported rating of the availability of external finance as an obstacle to business. We treat firms that rate finance as more of an obstacle as indicative of having lower financial access.[5] Cumulative national expenditures on completed energy and telecommunications investments are measured in the year each firm is surveyed, and are

[5]Since the WBES measure of financial access comes from the firm's self-reported rating of whether obtaining finance represents "an obstacle to business," we assume that firms rating finance as a greater obstacle have less financial access. This variable has a much higher response rate than more direct questions about access to finance such as the need for collateral when securing a loan. Kuntchev et al. (2013) show that these self-reported measures in the WBES are highly correlated with more direct measures of financial constraints.

converted to constant 2005 per capita U.S. dollars based on the year of completion. The third column in Table 1 reports the size of each sample.

It is striking that the average firm in the energy sample experiences nine power outages per month, and the loss of more than 24 hours of productive time on account of these outages underscores the potential importance of mitigating electrical disturbances. The fact that 43 percent of firms in the energy sample attempt to mitigate power instabilities by owning an electrical generator is also suggestive of the economic costs of lost production, and motivates our consideration of factors such as financial constraints that might impose barriers to generator purchases. Given how important the internet and websites have become for global sales and management of order flow, examination of factors that might also be holding website usage rates down to only 43 percent seems just as important. Thus, the descriptive statistics in Table 1 point to questions at the core of our analysis.

[Fig. 1 here]

Fig. 1 plots the linear relation between energy and telecommunication investments at the national level measured in logs of constant 2005 U.S. dollars per capita. The upper panel includes centered four-year moving averages of all available annual WDI observations from 2002 to 2014 among the 55 countries for which we have at least one observation for both types of national infrastructure expenditure, while the lower panel averages the available annual observations for each of the 55 countries. The important takeaway is that the two channels for the adoption of new technologies appear to be mutually reinforcing in the sense that countries with high levels of energy investment per capita also tend to make large investments in telecommunications technologies. If improvements in financial access can accelerate the adoption of both technologies, the benefits would clearly be two-fold.

[Fig. 2 here]

The data also allow us to consider the differential impact of infrastructure spending and financial access across firm sizes. Fig. 2 shows the frequency distributions of power outages and their durations by size measured as the number of employees. Although the distributions have similar shapes, the distinguishing features are the increased incidence and longer average durations of power outages for smaller firms relative to larger ones, which result in higher total outage hours for smaller firms. This suggests that small firms might benefit most from a reduction in power outages. Fig. 3 shows the log distribution of sales by firm size and ownership type. As might be expected, sales are increasing in firm size, but there remains a significant overlap in sales across firm sizes (i.e., many small firms as measured by the number of employees have higher sales than large firms). Thus, when examining firm-level growth we will consider it in terms of both sales and the number of employees. The distribution of firm sales appears to be bimodal, with a majority of firms having sales that are larger than average. Government firms comprise less than six percent of the sample, and are on average smaller and have a less peaked distribution of sales than private firms.

[Fig. 3 here]

3. Financial access and the efficiency of infrastructure investment

3.1. Empirical strategy

Our main strategy is to regress firm-level outcomes reflecting the application and effectiveness of electrical and telecommunications technologies on the WDI measures of countrywide infrastructure spending in these areas. Because we observe cumulative infrastructure spending for projects only in the year during which they come online, we might expect the completion of large electrification and telecommunications projects, which could take several years to finish, to relate with decreases in the number of hours of power outages experienced by firms surveyed in that year. We also interact cumulative expenditure on completed infrastructure projects with financial access at the firm level to test whether firms with greater financial access see near-term increases in the use of electrical and telecommunications applications. Because we include country indicator variables, the specifications aim to capture within-country variations in firm-level outcomes. The full specification is given by

$$
\begin{aligned}
ln(InfrastructureQuality_{i,t}) \quad = \quad & \beta_1 ln(Investment_{c,t}) + \beta_2 FinancialAccess_{i,t} \\
& +\beta_3 ln(Investment_{c,t}) \times FinancialAccess_{i,t} \\
& +\beta_4 SmallFirm_{i,t} + \beta_5 MediumFirm_{i,t} + \beta_6 Government_{i,t} \\
& +\beta_7 Manufacturing_{i,t} + \beta_8 Services_{i,t} + \beta_9 Exporter_{i,t} \\
& +\gamma_c + \nu_t + \varepsilon_{i,t}, \qquad (1)
\end{aligned}
$$

where $InfrastructureQuality_{i,t}$ represents a firm-level measure of the quality of infrastructure such as the log of the average number of power outages experienced in a month. $Investment_{c,t}$ is the log of per capita investment at the country level during year t in the relevant type of infrastructure. We also include $FinancialAccess_{i,t}$, which is the self-reported measure of access to finance at the firm level. The interaction $ln(Investment_{c,t}) \times FinancialAccess_{i,t}$ captures whether financial access is important for enabling firms to gain immediately from completed countrywide spending on infrastructure. A finding that firms with greater access to finance achieve larger benefits from recent infrastructure spending would support our hypothesis that financial access is a key factor in technological adoptions. For notational convenience, we use the subscript c for country-level variables and i for firm-level variables throughout. For example, from Eq. 1, $Investment_{c,t}$ and $FinancialAccess_{i,t}$ are measured at the country and firm levels, respectively.

We also include a number of controls, both at the firm and country levels, in the baseline specification. At the firm level, the binary indicator variables $SmallFirm$ and $MediumFirm$ control for firm size. The WBES defines a small firm as one with fewer than 20 employees, a medium firm as one with between 20 and 99 employees, and a large firm with 100 or more. The dummy variable $Government$ takes a value of unity if the firm is majority-owned by the government and zero otherwise. We control for government ownership as this may affect a firm's ability to gain access to finance. The binary indicators $Manufacturing$ and $Services$ control for the firm's basic industry, while *Exporter* takes a value of unity if the firm exports at least some portion of its product and zero otherwise. We include this final control because firms that export a portion of their products may operate differently than their domestic counterparts due to the need to satisfy regulatory requirements in both domestic and foreign markets.

Our specification is structured to examine whether firms with more financial access benefit more in the immediate term from completed infrastructure projects. Our main identifying assumption is that financial access is not correlated with other firm characteristics which would help them disproportionally benefit from country-level infrastructure investment. This is clearly a strong assumption. For example, firms located in urban areas are likely to experience higher levels of financial access and also may be more situated to take advantage of completed infrastructure projects. More generally, financial access may correlate with other types of access that predict a differential impact from infrastructure projects. We attempt to provide a test for this in an alternate specification in which we replace our financial access variable with a similarly constructed transportation access variable. This variable should capture many of the same non-financial factors that might predict a firm being positioned to take advantage of infrastructure projects. If firms with more transportation access do not differentially benefit from infrastructure, however (as Table 4 indeed indicates), this would bolster support for our effect being in fact specific to financial access.

Country and year-specific fixed effects also appear in the baseline. The country fixed effects are important because, among many other omitted factors, they control for the overall level of infrastructure in each country, allowing us to compare the effects of infrastructure spending within countries. Since it is possible that countries with more severe problems with power outages, for example, spend more on electricity infrastructure, controlling for the overall level of electrical infrastructure ameliorates some of the effects of simultaneity in our analysis. Because the sampling methodology of the WBES has changed over time, the regressions also utilize the sampling weights provided in the WBES data.

3.2. Financial access and energy infrastructure

The regressions reported in Table 2 focus on the "energy sample" and explore the effects of countrywide spending on energy infrastructure and firm-level financial access on power disruptions. The dependent variable in columns 1 and 2 is the log of the total number of power outage hours experienced by a firm in a typical month. This is calculated from the WBES data as the product of the average number of monthly outages during the survey year and the average outage duration, and is the most relevant single measure of power outages available in the data. For the specifications reported in columns 3-6, we break down total hours into its components, which allows us to determine whether the log of the number of outages or the log of their average duration is the more responsive one. Columns 2, 4, and 6 include interactions of firm-level financial access with the log of national infrastructure expenditure per capita.

[Table 2 here]

As expected, higher levels of investment in energy infrastructure in a country are associated with fewer overall hours of power outages for firms therein (columns 1 and 2) and also with shorter and less frequent outages (columns 3-6). When financial access enters the specification independently it is consistently associated with fewer and shorter outages. In addition, the interaction between infrastructure spending and financial access indicates that firms with more financial access see larger immediate reductions in outages. We take this as evidence that financial access is a key component in taking advantage of recently completed country-level investments.

Although we find a strongly statistically significant relationship between the level of financial access of the firm and the size of the impact, the magnitude of the coefficients in Table 2 do not appear large. We believe that this is primarily due to the structure of our data, in which the country-level infrastructure spending is likely not affecting all firms in our sample. An infrastructure project completed in a particular region may not have a significant impact on firms outside that region. Unfortunately, our data do not allow us to determine the particular location of firms in relation to completed projects.

The effect of this will be a significant number of firms included in our data which may be unaffected by completed infrastructure projects, diluting the measured impact of these projects and diminishing the size of the coefficients. We believe it likely that if we were able to link the particular location of firms to the location of completed investments, the size of our coefficients would increase substantially.[6]

Even if we take the coefficients in Table 2 at face value, however, they nevertheless indicate substantive cumulative effects of financial access on firm-level outages. According to the specification in column 2, even after controlling for country fixed effects, a one standard deviation increase of 1.4 in the financial access measure from its sample mean of 2.4, with no change in energy investment, implies a reduction in monthly outage hours of an average firm from 24 to 21.98 hours. Add to this a one standard deviation increase in the log of energy investment from its cross-country mean of $31.67 to $116.76, and monthly outage hours fall from 24 to 19.64 hours. These figures account to reductions in annual outage hours per firm of 24.2 and 52.3 hours, respectively. Given an average workforce size of 18.6 million persons for the WDI countries in our sample and an average manufacturing share in GDP of 22.3%, this implies a savings of more than 100 million man-hours per year (18.6*.223*24.2) for the change in financial access alone and nearly 217 million man hours for the combined change in energy investment and financial access. At a PPP-adjusted wage of $12 per hour, these savings add up to $1.2 billion and $2.6 billion respectively for an average country in the sample. These numbers suggest that the relaxation of financial constraints can be a key factor in reducing the number of productive hours lost to power outages, and in doing so enhancing the effectiveness of infrastructure spending.

Turning to the other variables, the results indicate that small and medium size firms experience significantly longer power outages on average. However, due to a negative

[6] Although our data do not match the locations of infrastructure investments with nearby firms, they do indicate whether a firm is located in a large city (population of more than one million), and thus we can gauge the impact of infrastructure investment on firms outside of these large, urban environments. Table A4 in Appendix B shows that when we replace the interaction term in Table 2 with one denoting firms outside of large cities, our electrification coefficients increase slightly while all remain statistically significant at the 5 percent level. This additional finding sheds some light on how the impact of these investments are widespread and, more importantly, not limited to the narrow locales where the infrastructure projects are headquartered.

coefficient on the number of outages for each type, they do not experience a significantly higher number of total outage hours than large firms. Although large firms are more likely to have the resources to mitigate power outages, they also likely demand more electricity from the grid. However, we see that government-owned firms experience significantly shorter power outages compared to private firms, leading to a large and significant difference in their overall outage hours. One explanation for this result could be that government firms are more likely to have strong political connections and thus experience a more prompt response in the case of an outage.

[Table 3 here]

One of the assumptions made in our main specification is that we can detect an impact from the electrical investment in the same year in which it is completed. To test the importance of this assumption, we use several different specifications for the timing of the electrical investment in Table 3. In the first two columns, we replace our measure of electrical investment with the average investment over the past two years, while in the last two columns we use the average over the past three years. The results are robust to both of these alternate specifications.

[Table 4 here]

In Table 4, we replace our financial access variable with a similarly constructed transportation access variable. This is designed to test whether our positive results above are due to financial access being correlated with other types of access to infrastructure investments. As we would expect, transportation access by itself is negatively associated with power outages. The interaction term, however, is not significant. Thus, firms with more transportation access do not appear to take better advantage of infrastructure investments.

In light of these results, it is natural to consider channels through which financial access could magnify the effects of energy spending. One possibility is that greater financial access enables firms to invest in generators that improve the stability of their power source, and that these generators act as a complement to the power grid rather than a subsitute. In the energy sample, 43 percent of firms report owning a generator, and yet among these firms only five percent derive the majority of their power from a generator. This suggests that the vast majority of firms use generators in a backup role to complement the power grid. Indeed, as we see in Fig. 2, the largest firms appear to be the most energy-constrained.

If generators and the central power infrastructure are complementary, then companies that can afford to invest in a generator will be better able to take advantage of general improvements in energy infrastructure. In an examination of firms in Sub-Saharan Africa, Steinbuks (2011) uses the WBES to show that increased financial access leads firms to invest more in generators when the quality of their power grid is poor. We investigate the more general relationship in a broader set of countries in Table 5, which reports coefficients from probit regressions that capture the impact of financial access on generator ownership. Marginal effects evaluated at the means are shown in brackets. The dependent variable is whether or not the business owns a generator, and the main variable of interest is its level of financial access. Columns 3-5 break the sample into small, medium, and large firms. Improved financial access makes generator ownership significantly more likely, which implies that as energy infrastructure improves, firms that have greater levels of financial access are better able to invest in the complementary technology of generators, and therefore can take better advantage of the energy investment. This effect is remarkably consistent across firms of different sizes; small firms are able to use financial access to obtain a generator just as easily as medium and large firms.

[Fig. 4 here]

To further illustrate the magnitude of the impact from our regression results, in Fig. 4 we plot the total amount of investments in energy infrastructure times the average firm-level measure of financial access (i.e., the interaction term between energy investments and financial access in our specification) for 14 nations in Sub-Saharan Africa. The key insight is that countries with higher interaction terms tend to have lower outage hours (i.e., brighter "red" shades in the upper panel of the figure tend to correspond with softer shades in the lower panel, and vice versa); that is, the combination of a high level of investment in electricity capacity and the provision of better financial access implies lower outages hours in these countries. Cameroon, Liberia, and South Africa offer strong examples of this relationship.

3.3. Financial access and telecommunications infrastructure

The WDI dataset also provides information on countrywide investments in telecommunications infrastructure on a cumulative basis for projects completed in a given year. To test the relationship between financial access and telecommunications investments, we run the specification in Eq. 3 for telecommunications rather than energy investment. Table 6 examines the impact of these variables on the frequency of website usage at the firm level. The dependent variable, $WebsiteUse$, takes a value of unity if the firm uses a website for business purposes and zero otherwise. We focus on website usage due to its high response rate in the survey and because it is likely that some level of investment by the firm is necessary to take advantage of the general availability of the internet. Marginal effects of the probit regressions evaluated at the means are shown in brackets.

[Table 6 here]

As expected, higher levels of spending on telecommunications infrastructure are associated with higher rates of website usage by firms when the interaction term is excluded. Moreover, financial access when entered independently into the specification has a statistically significant effect on website usage. Finally, the interaction term indicates that financial access is a critical determinant of the impact of telecommunications spending itself. When the interaction term is included, the level of investment in that year is no longer significant on its own; it only improves outcomes in conjunction with better financial access. We take this to mean that financial access is central to facilitating firms' use of websites. Rather than affecting all firms equally, spending on telecommunications by the national government is immediately beneficial only for firms that have the financial resources to make complementary investments in telecommunications technology. Firms without financial access may eventually benefit, but likely do not in the year the investment is completed.

The remaining variables indicate that, as we might expect, small and medium size firms are significantly less likely to have a business website than large firms, with small firms being the least likely. Interestingly, government-owned firms are less likely than private firms to have a business website, contrasting with the previous result that government-owned firms experience fewer hours of power outages than private firms. This could suggest that government firms are, on average, less sophisticated or technologically savvy than private firms.[7]

4. Implications for firm-level growth

In Section 3, we demonstrated that financial access improves the effectiveness of infrastructure spending in relaxing capacity or technological constraints on firms. Here, we explore the relationship between the severity of these constraints and firm growth. In Eq. 2, we attempt to capture the relationship between the quality of infrastructure and growth in firm sales and employment. The dependent variable is the cumulative growth in either sales or employees over the course of the three years prior to the survey, while the main explanatory variables of interest are our measures of outage hours and website use in the most recent fiscal year. This empirical strategy assumes some level of stability in outage hours and website use across the three years prior to the survey. That is, if a firm had a large number of outage hours in the most recent year, they likely had similar issues over the previous two, and we can capture the relationship between those outages and firm growth over that period. This empirical specification is similar to that found in Beck et al. (2005), which uses data with the same structure to capture relationships between firm growth and institutions (and particularly financial development). Equation 2 presents our full specification for investments in electrical and telecommunications infrastructure:

[7] Table A5 in Appendix B shows that when we replace our interaction term with one focusing only on firms outside of large cities, the investment coefficients are essentially unchanged, suggesting that, as shown earlier for electricity infrastructure, the impact of aggregated telecommunications investments may be far-reaching within a country.

$$
\begin{aligned}
FirmGrowth_{i,t} \;=\; & \beta_1 ln(OutageHours)_{i,t} + \beta_2 WebsiteUse_{i,t} + \beta_3 FinancialAccess_{i,t} \\
& + \beta_4 OwnGenerator_{i,t} + \beta_5 ln(InitialLevelofDependentVariable_{i,t}) \\
& + \beta_6 SmallFirm_{i,t} + \beta_7 MediumFirm_{i,t} + \beta_8 Government_{i,t} \\
& + \beta_9 Manufacturing_{i,t} + \beta_{10} Services_{i,t} \\
& + \beta_{11} Exporter_{i,t} + \gamma_c + \nu_t + \varepsilon_{i,t}. \qquad (2)
\end{aligned}
$$

Most of the independent variables in Eq. 2 are specified as they were in Section 3. The log of the initial level of sales or employees at the start of the three-year period is included to control for a possible convergence or catch-up effect. The main explanatory variables of interest are $ln(OutageHours)_{i,t}$ and $WebsiteUse_{i,t}$. We also include financial access and ownership of a generator to control for the direct effects of these factors.

[Table 7 here]

The results for both technologies are shown in Table 7. The central finding is that firms facing more hours of power outages tend to exhibit slower growth in terms of both sales and the number of employees, while those able to use websites see stronger growth along both dimensions. The results for website usage are strong and robust. The coefficients on outage hours are significant at the ten percent level in columns 1-4, and are just outside the ten percent level of significance when the size controls are included in the final two columns. We take these results to indicate that outages are generally associated with lower rates of firm growth. In addition to these findings, we observe the strong positive relationship between financial access and growth that Beck et al. (2005) found in their work. As we might expect, we also find that firms with access to a generator tend to grow faster. Finally, we see a strong convergence effect in both sales and employees.

In Section 3, we demonstrated that financial access enhances the effectiveness of infrastructure investment in decreasing the prevalence of power outages and in expanding access to websites. Here we have shown that both channels are positively related to growth in sales and employees at the firm level. These findings suggest that the effectiveness of infrastructure spending provides an alternative channel through which financial development can promote real economic activity.

5. Conclusion

We investigate the link between financial development at the firm level and the expansion of electrification and telecommunication technologies in promoting firm growth. Using firm-level surveys from 2002-2014, we find that firms with better access to finance can more fully capitalize on country-level spending on electrical and telecommunications infrastructure. Further, we find that these two technologies play an important role in firm growth. Our findings highlight the role of financial development in determining the effectiveness of improvements in energy infrastructure. The level of access to finance that a firm has can help to determine its ability to benefit from energy infrastructure projects, and translate better infrastructure into improved economic outcomes. This result offers an alternative channel for finance to affect growth in a development setting, suggesting that the complementarity of financial development and energy infrastructure can be important.

Our findings have several policy implications. First and foremost, governments may improve real-sector outcomes by recognizing that access to finance at the firm level plays a key role in determining the effectiveness of infrastructure expansion. Nations seeking to improve their infrastructures should note the areas where complementary investments by firms are needed, and focus on improving firm access to finance to facilitate these investments. Second, attempts to estimate the effectiveness of a proposed amount of infrastructure spending should consider the level of financial access for the firms in question, and recognize that spending focused on firms with greater financial access could be the most effective. Finally, our findings reinforce the importance of infrastructure spending on electrification and telecommunications by illustrating the link between these technologies and firm growth. Many countries still face significant limitations in the extent of their electrical and telecommunications infrastructures, and improving these conditions could have profound effects.

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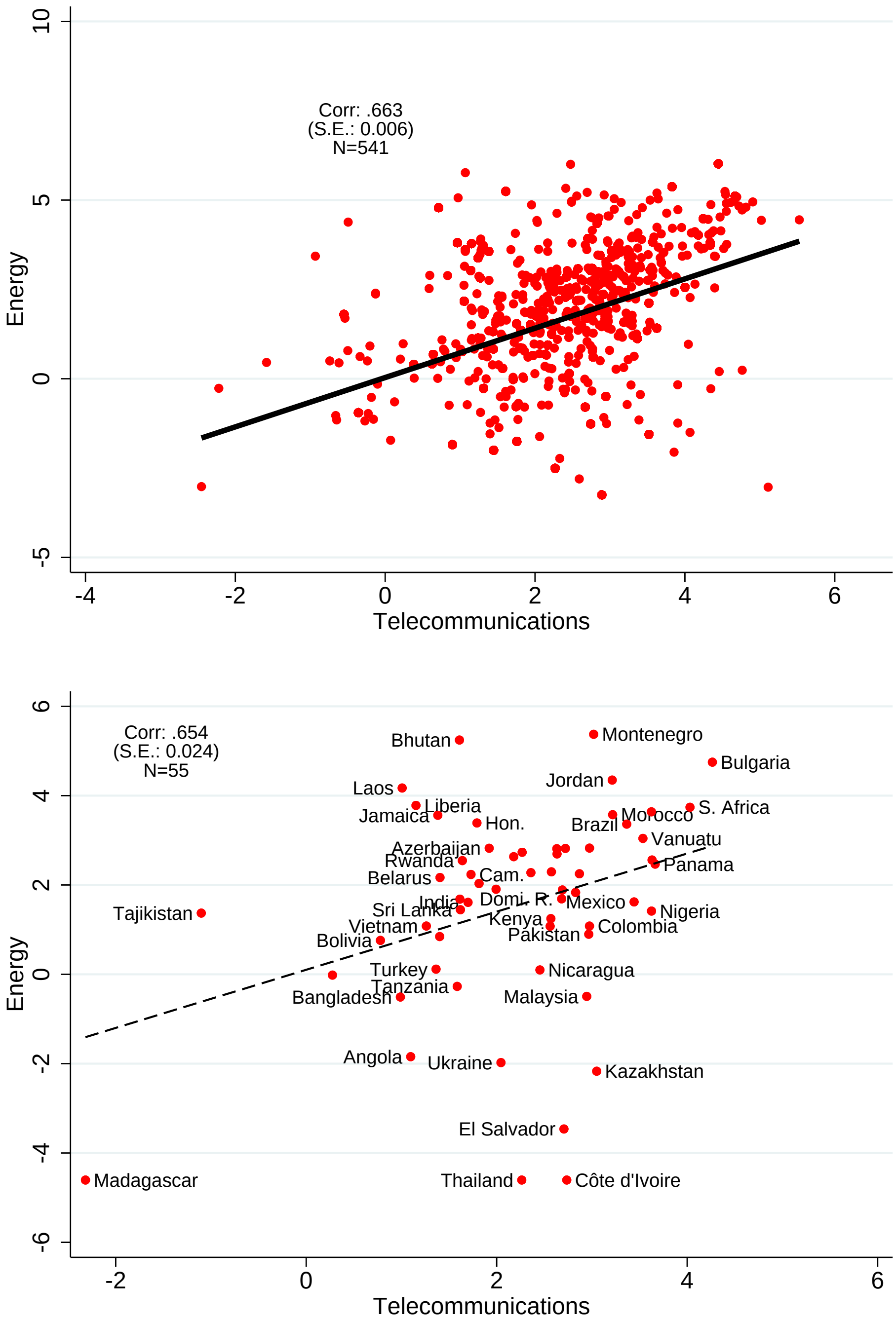

Fig. 1. Investments in energy and telecommunication infrastructure.

Note: The upper panel plots the relation between the logs of the four-year moving averages of energy and telecommunication investments in constant 2005 U.S. dollars per capita for the 55 countries common to the "energy" and "telecommunications" samples. The lower panel plots the investment data with one observation per country by averaging across all available years.

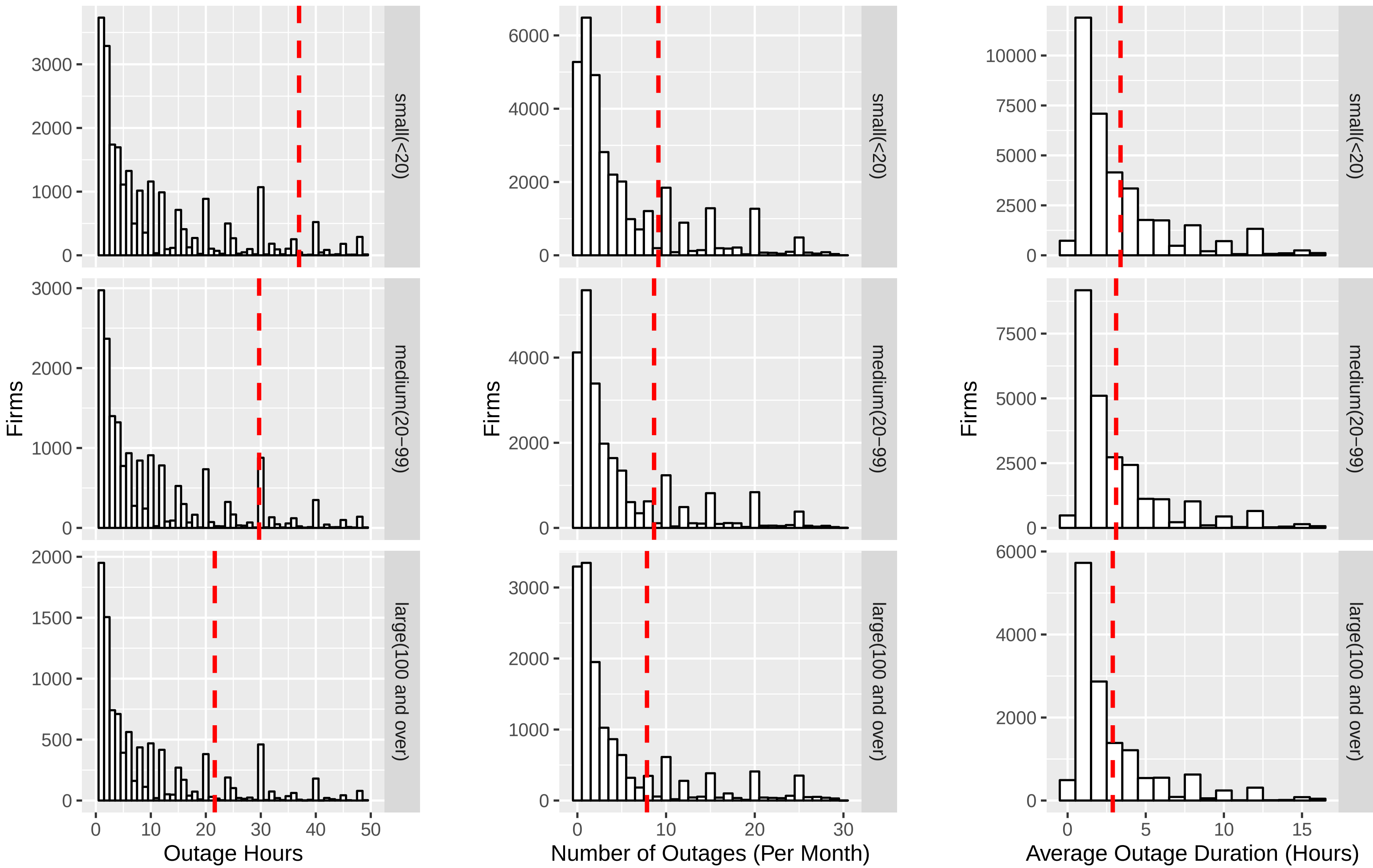

Fig. 2. Outages and their duration by firm size.

Note: Fig. 1 plots histograms of the three main electrical outcome variables in the "energy sample" by firm size as measured by the number of employees. The unit for "Outage Hours" is average monthly hours per firm surveyed. "Outage Duration (Hours)" is defined as the mean duration, in hours, of an outage incident for each surveyed firm.

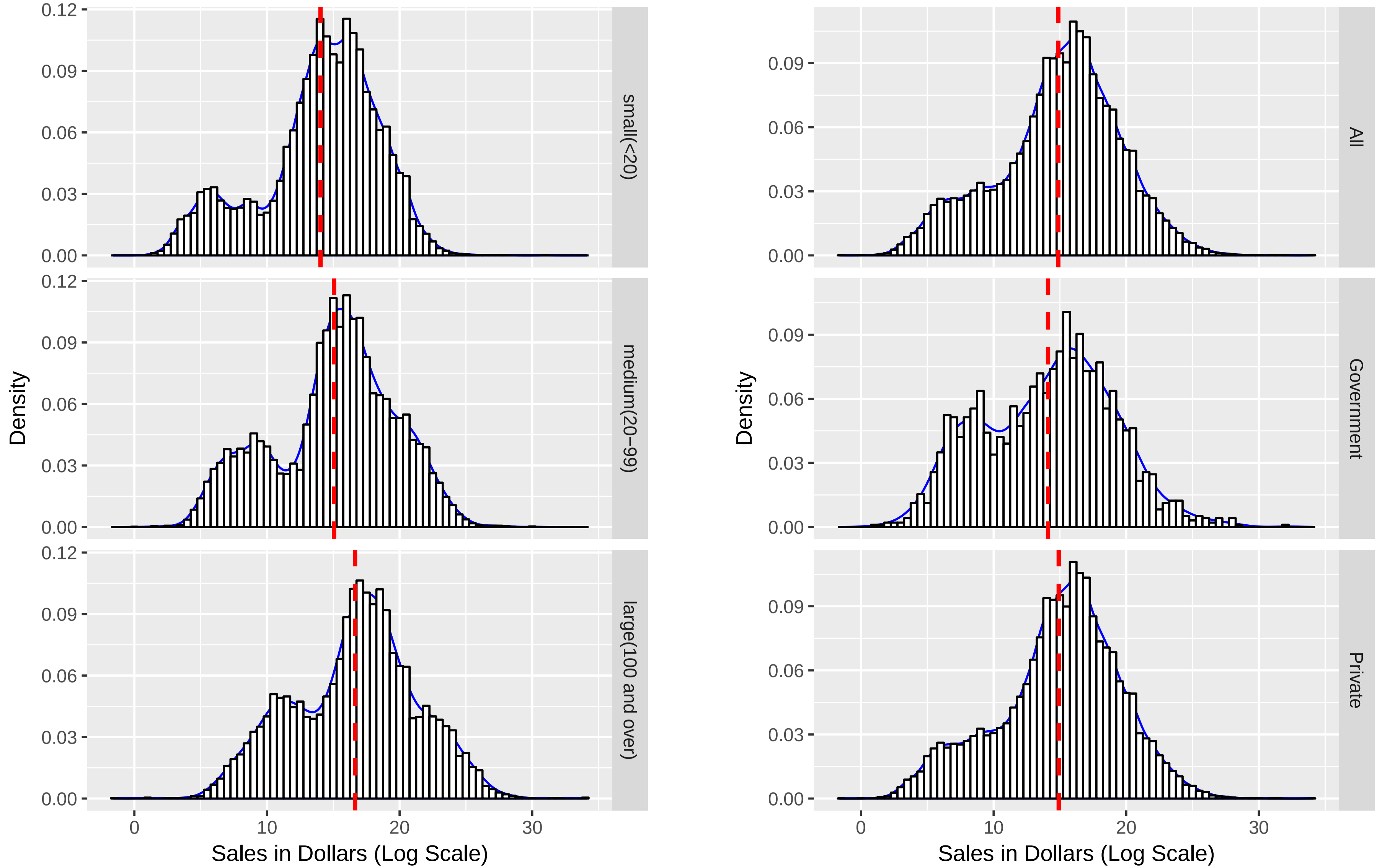

Fig. 3. Distribution of sales in logs by the number of employees and sector.

Note: In constant U.S. dollars (2005). Here we restrict the sample size to the observations that we use in our regression. The regression sample is defined as the full sample of firm-level survey variable for firms in countries that we have both observations for electricity and telecommunication investments.

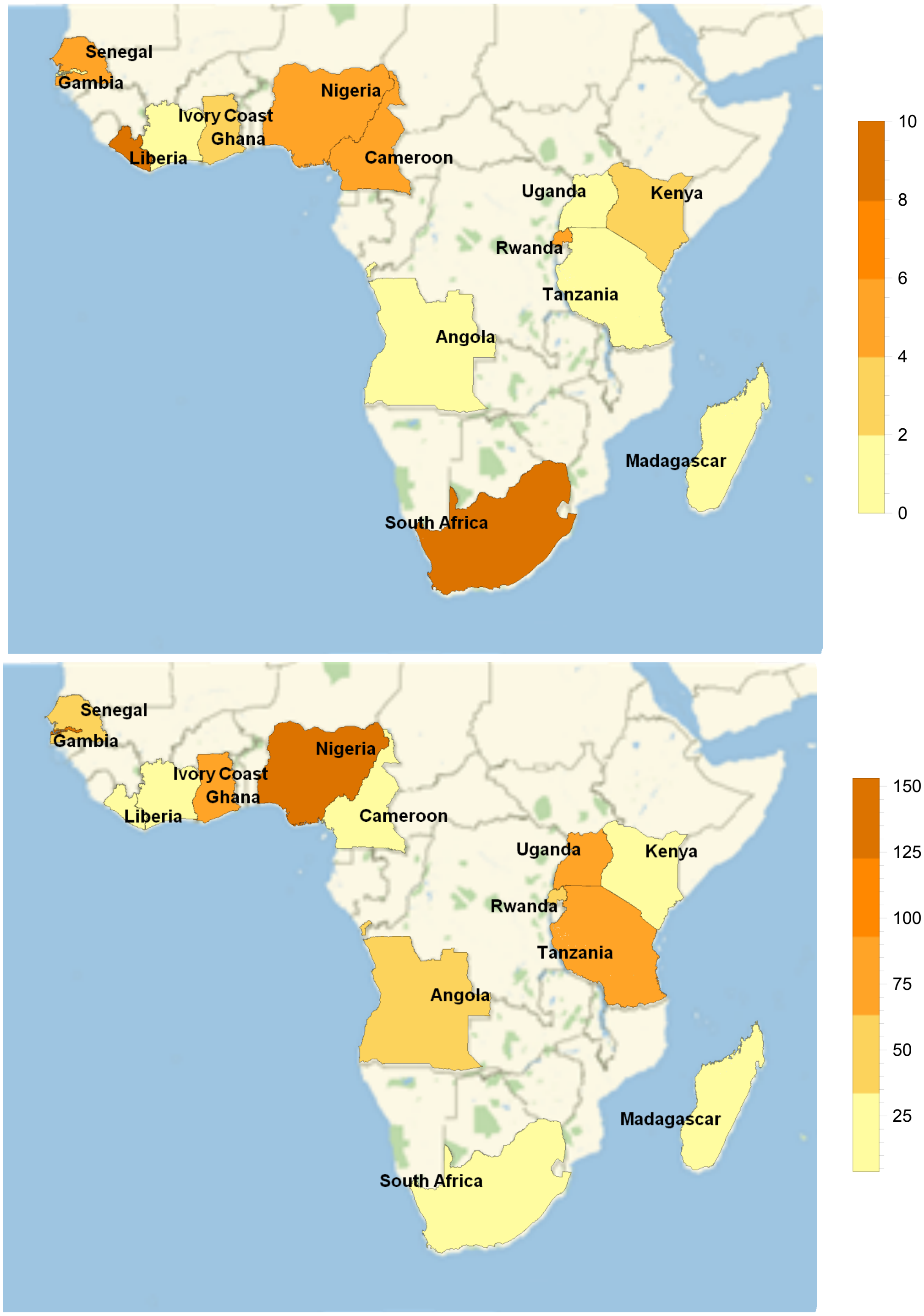

Fig. 4. Outage hours and the interactions of energy investment with financial access for 14 countries in sub-Saharan Africa.

Note: The figure in the upper panel plots the log of national investments in energy infrastructure times financial access for selected African countries in the "energy sample," and the figure in the lower panel plots the corresponding number of outage hours. Data are averaged across firms and years.

Table 1. Descriptive statistics.

Variable	Mean	Std. Dev.	N
Firm-Level Variables			
Energy Sample			
Number of Outages per Firm	9.0	13.6	32,542
Outage Duration (Hours)	3.0	3.0	32,542
Outage Hours per Firm	24.0	53.2	32,542
Financial Access (Firm-Level)	2.4	1.4	32,542
Generator Ownership (% of Firms)	43.0	49.5	26,887
Telecommunications Sample			
Website Use (% of Firms)	43.1	49.9	115,788
Financial Access (Firm-Level)	2.3	1.4	115,788
Country-Level Variables			
Energy Investment Per Capita	$ 31.67	$ 85.09	873
Telecom Investment Per Capita	$ 20.82	$ 81.00	1,979

Note: The table reports summary statistics for the two main samples used in our regressions: the "energy sample" and the "telecommunications sample." These samples reflect the availability of national observations for investment in electrical and telecommunications infrastructure coupled with the availability of the relevant firm-level data within these countries. "Energy Investment per capita" and "Telecom Investment per capita" are pooled averages across years in constant 2005 U.S. dollars. "Financial Access" is a count variable ranging from 0-4, with 4 representing the highest level of financial access. The unit for "Outage Hours" is the pooled average of total hours per year per firm surveyed. "Outage Duration (Hours)" is defined as the duration, in hours, of an average outage incident. "Website Use" is a binary variable set to unity for firms using a website and zero otherwise.

Table 2. The effects of investments in energy infrastructure and financial access on electrical outages (with full set of controls).

	Outage Hours		Outage Duration		Number of Outages	
	(1)	(2)	(3)	(4)	(5)	(6)
Electrical	-0.072***	-0.054***	-0.021***	-0.013**	-0.051***	-0.040***
Investment (E)	(0.008)	(0.010)	(0.005)	(0.006)	(0.006)	(0.008)
Financial Access (F)	-0.040***	-0.033***	-0.017***	-0.014***	-0.023***	-0.019***
	(0.005)	(0.006)	(0.003)	(0.004)	(0.004)	(0.005)
ExF		-0.008***		-0.003**		-0.005***
		(0.002)		(0.001)		(0.002)
Small	0.020	0.021	0.055***	0.055***	-0.036**	-0.036**
	(0.019)	(0.019)	(0.012)	(0.012)	(0.016)	(0.016)
Medium	0.019	0.020	0.038***	0.038***	-0.020	-0.020
	(0.018)	(0.018)	(0.011)	(0.011)	(0.014)	(0.014)
Government	-0.139**	-0.138**	-0.111***	-0.110***	-0.048	-0.047
	(0.054)	(0.054)	(0.035)	(0.035)	(0.040)	(0.040)
Manufacturing	0.096***	0.097***	0.072***	0.072***	0.022	0.023
	(0.020)	(0.020)	(0.012)	(0.012)	(0.016)	(0.016)
Services	-0.007	-0.007	-0.012	-0.012	0.003	0.003
	(0.024)	(0.024)	(0.014)	(0.014)	(0.019)	(0.019)
Exporter	-0.021	-0.021	0.002	0.002	-0.023*	-0.023*
	(0.017)	(0.017)	(0.010)	(0.010)	(0.013)	(0.013)
Country Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	32,542	32,542	32,542	32,542	32,542	32,542
R^2	0.591	0.591	0.284	0.285	0.685	0.685

Note: The regression estimates reported in the table are from the following specification: $ln(InfrastructureQuality_{i,t}) = \beta_0 + \beta_1 ln(Investment_{c,t}) + \beta_2 FinancialAccess_{i,t} + \beta_3 ln(Investment_{c,t}) \times FinancialAccess_{i,t} + \varepsilon_{i,t}$. We also include dummy variables for countries and years. $FinancialAccess$ is coded on a scale from 0-4 and is increasing in access. The sample is restricted to firms for which full outage and national investment in energy infrastructure are available. Standard errors are in parentheses beneath the coefficient estimates. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.

Table 3. The effects of investments in energy infrastructure and financial access on electrical outages (with alternate timing of investment measurement).

	Dependent variable: Log of Outage Hours			
	(1)	(2)	(3)	(4)
	Investment averaged over 2 years		Investment averaged over 3 years	
Electrical Investment	-0.087***	-0.071***		
(E, 2 year average)	(0.008)	(0.011)		
Electrical Investment			-0.077***	-0.067***
(E, 3 year average)			(0.007)	(0.009)
Financial Access (F)	-0.041***	-0.031***	-0.041***	-0.037***
	(0.005)	(0.007)	(0.005)	(0.006)
ExF (2 year average)		-0.007**		
		(0.003)		
ExF (3 year average)				-0.004**
				(0.002)
Small	-0.003	-0.003	-0.002	-0.002
	(0.016)	(0.016)	(0.016)	(0.016)
Medium	-0.025	-0.025	-0.023	-0.024
	(0.019)	(0.019)	(0.019)	(0.019)
Government	-0.120**	-0.119**	-0.129**	-0.129**
	(0.054)	(0.054)	(0.054)	(0.054)
Manufacturing	0.101***	0.101***	0.099***	0.099***
	(0.020)	(0.020)	(0.020)	(0.020)
Services	-0.005	-0.006	-0.006	-0.006
	(0.024)	(0.024)	(0.024)	(0.024)
Exporter	-0.020	-0.020	-0.021	-0.021
	(0.017)	(0.017)	(0.017)	(0.017)
Country Dummies	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes
Observations	32,542	32,542	32,542	32,542
R^2	0.591	0.591	0.591	0.591

Note: The regression estimates reported in the table are from the following specification: $ln(InfrastructureQuality_i) = \beta_1 ln(Investment_{c,t}) + \beta_2 FinancialAccess_{i,t} + \beta_3 ln(Investment_{c,t}) * FinancialAccess_{i,t} + \beta_4 SmallFirm_{i,t} + \beta_5 MediumFirm_{i,t} + \beta_6 Ownership_{i,t} + \beta_7 Manufacturing_{i,t} + \beta_8 Services_{i,t} + \beta_9 Exporter_{i,t} + \varepsilon$. We also include dummy variables for countries and years. Investment here is averaged over either the previous two or three years, depending on the regression. $FinancialAccess$ is coded on a scale from 0-4 and is increasing in access. The sample is restricted to firms for which full outage and national investment in energy infrastructure are available. Standard errors are in parentheses beneath the coefficient estimates. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.

Table 4. The effects of investments in energy infrastructure and transportation access on electrical outages.

	Outage Hours		Outage Duration		Number of Outages	
	(1)	(2)	(3)	(4)	(5)	(6)
Electrical	-0.073***	-0.073***	-0.020***	-0.014**	-0.052***	-0.059***
Investment (E)	(0.008)	(0.011)	(0.005)	(0.007)	(0.007)	(0.009)
Transportation	-0.065***	-0.065***	-0.021***	-0.019***	-0.044***	-0.046***
Access (T)	(0.005)	(0.006)	(0.003)	(0.004)	(0.004)	(0.005)
ExT		0.000		-0.002		0.002
		(0.002)		(0.001)		(0.002)
Small	-0.010	-0.010	-0.020**	-0.020**	0.011	0.011
	(0.016)	(0.016)	(0.009)	(0.009)	(0.013)	(0.013)
Medium	-0.041**	-0.041**	-0.063***	-0.063***	0.024	0.024
	(0.019)	(0.019)	(0.012)	(0.012)	(0.016)	(0.016)
Government	-0.143***	-0.143***	-0.112***	-0.112***	-0.051	-0.051
	(0.055)	(0.055)	(0.036)	(0.036)	(0.040)	(0.040)
Manufacturing	0.094***	0.094***	0.073***	0.073***	0.019	0.019
	(0.020)	(0.020)	(0.012)	(0.012)	(0.016)	(0.016)
Services	-0.008	-0.008	-0.013	-0.013	0.002	0.002
	(0.025)	(0.025)	(0.014)	(0.014)	(0.019)	(0.019)
Exporter	-0.026	-0.026	-0.000	-0.000	-0.026*	-0.025*
	(0.017)	(0.017)	(0.010)	(0.010)	(0.013)	(0.013)
Country Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	32,241	32,241	32,241	32,241	32,241	32,241
R^2	0.593	0.593	0.285	0.285	0.686	0.686

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The regression estimates reported in the table are from the following specification: $ln(InfrastructureQuality_i) = \beta_1 ln(Investment_{c,t}) + \beta_2 TransportationAccess_{i,t} + \beta_3 ln(Investment_{c,t}) * TransportationAccess_{i,t} + \beta_4 SmallFirm_{i,t} + \beta_5 MediumFirm_{i,t} + \beta_6 Ownership_{i,t} + \beta_7 Manufacturing_{i,t} + \beta_8 Services_{i,t} + \beta_9 Exporter_{i,t} + \varepsilon$. We also include dummy variables for countries and years. Investment here is averaged over either the previous two or three years, depending on the regression. $TransportationAccess$ is coded on a scale from 0-4 and is increasing in access. The sample is restricted to firms for which full outage and national investment in energy infrastructure are available. Standard errors are in parentheses beneath the coefficient estimates. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.

Table 5. Probit estimates of the effects of financial access on generator ownership.

	Owning a Generator				
	(1)	(2)	(3)	(4)	(5)
	All		Small	Medium	Large
Financial Access	0.069***	0.046***	0.046***	0.042***	0.042***
	0.027***	0.018***	0.015***	0.017***	0.016***
	[0.007]	[0.007]	[0.012]	[0.011]	[0.014]
All Controls	No	Yes	Yes	Yes	Yes
Country Dummies	Yes	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes	Yes
Observations	26,887	26,887	9,440	10,190	7,124
$Pseudo-R^2$	0.186	0.268	0.186	0.275	0.265

Note: The results reported in Column 1 of the table do not include the additional control set, while those reported in Columns 2-6 include the full set from Eq. 1 (i.e., dummy variables for firm size, ownership type, sector, and exporter status, as well as GDP per capita and the level of financial development as measured by the national ratio of M2 to GDP). $FinancialAccess$ is coded on a scale from 0-4 and is increasing in access. The sample is restricted to firms for which full outage and national investment in energy infrastructure are available, as in the "energy sample" of Tables 2-3, but also to only those firms who responded to the survey question about owning a generator. Marginal effects are shown in brackets, and standard errors are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.

Table 6. Probit estimates of the effects of investment and financial access on website use.

	Website Use			
	(1)	(2)	(3)	(4)
Telecommunication	0.005	-0.004	0.012***	0.003
Investment (T)	(0.004)	(0.005)	(0.004)	(0.005)
	[0.002]	[-0.001]	[0.005]	[0.001]
Financial Access (F)	0.042***	0.036***	0.027***	0.021***
	(0.003)	(0.004)	(0.003)	(0.004)
	[0.016]	[0.014]	[0.011]	[0.008]
TxF		0.004***		0.004***
		(0.001)		(0.001)
		[0.002]		[0.002]
Small			-1.031***	-1.031***
			(0.012)	(0.012)
Medium			-0.489***	-0.489***
			(0.012)	(0.012)
Government			-0.115***	-0.115***
			(0.023)	(0.023)
Manufacturing			-0.256***	-0.256***
			(0.012)	(0.012)
Services			-0.044***	-0.044***
			(0.014)	(0.014)
Exporter			0.508***	0.508***
			(0.010)	(0.010)
Country Dummies	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes
Observations	115,788	115,788	115,788	115,788
$Pseudo-R^2$	0.111	0.111	0.201	0.201

Note: Standard errors are in parentheses beneath the coefficient estimates, and marginal effects are in brackets. The probit regressions reported in the table include the full set of controls in Eq. 1 (i.e., dummy variables for firm size, ownership type, sector, and exporter status). $FinancialAccess$ is coded on a scale from 0-4 and is increasing in access. The sample is restricted to firms for which firm-level data on website usage and national data on investment in telecommunications infrastrcuture are available. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.

Table 7. The impact of outages and website usage on firm growth.

	(1) Sales	(2) Employees	(3) Sales	(4) Employees	(5) Sales	(6) Employees
ln(Outage Hours)	-0.013*	-0.003*	-0.013*	-0.003**	-0.014	-0.002
	(0.008)	(0.002)	(0.008)	(0.002)	(0.009)	(0.002)
Website Use	0.261***	0.082***	0.259***	0.082***	0.197***	0.059***
	(0.020)	(0.004)	(0.020)	(0.004)	(0.022)	(0.004)
Financial Access	0.021***	0.005***	0.020***	0.005***	0.016**	0.003**
	(0.007)	(0.001)	(0.007)	(0.001)	(0.008)	(0.001)
Own Generator	0.175***	0.056***	0.172***	0.057***	0.122***	0.038***
	(0.021)	(0.004)	(0.021)	(0.004)	(0.024)	(0.004)
ln(Initial Sales)	-0.151***		-0.155***		-0.259***	
	(0.006)		(0.006)		(0.009)	
ln(Initial Employees)		-0.084***		-0.086***		-0.326***
		(0.002)		(0.002)		(0.004)
Constant	2.205*	0.392***	2.334*	0.454***	5.206***	0.849***
	(1.323)	(0.128)	(1.335)	(0.129)	(1.369)	(0.126)
Size Controls	No	No	No	No	Yes	Yes
All Other Controls	No	No	Yes	Yes	Yes	Yes
Country Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	35,598	46,371	35,598	46,371	29,323	36,639
R^2	0.087	0.097	0.088	0.099	0.116	0.286

Note: The regression estimates reported in the table are from the following specification: $FirmGrowth_{i,t} = \beta_1 OutageObstacle_{i,t} + \beta_2 ln(InitialLevelofDependentVariable_{i,t}) + \beta_3 FinancialAccess_{i,t} * OutageObstacle_{i,t} + \beta_4 SmallFirm_{i,t} + \beta_5 MediumFirm_{i,t} + \beta_6 Ownership_{i,t} + \beta_7 Manufacturing_{i,t} + \beta_8 Services_{i,t} + \beta_9 Exporter_{i,t} + \varepsilon_{i,t}$, in which $FirmGrowth_{i,t}$ is measured by the growth rates of sales and the number of employees over the previous three year period. Standard errors are in parentheses beneath the coefficient estimates. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.

Appendices

Appendix A. Data

We use two main datasets for the project. First, we use the World Bank Enterprise Survey to obtain the firm level data. Second, we use the World Bank's World Development Indicators database to obtain macro-level data and control variables at the aggregate level. These latter variables are the real per capita rate of GDP growth and the level access to financial credit at the macro level.

The World Bank Enterprise Survey (WBES) is a firm-level survey that covers a broad range of topics related to access to finance, corruption, and other information related to conducting business in a wide range of countries. The survey is primarily answered by business owners and top managers of a sample of firms representative of the private sectors of the economies. Firms with 100 percent government/state ownership are not eligible for the survey. Firms are classified into small, medium, and large business across different manufacturing and service industries (classified by two-digit ISIC indices) across countries and regions as in Table A2.[8] We also present a detailed breakdown of the list of the countries in the sample in Table A3 by two samples: one used for the energy regressions and one used for the telecommunication regressions. Table A3 makes clear that our sample contains a remarkable level of detail across countries along with a relatively balanced coverage of small, medium, and large firms.

There two main waves of surveys: the first wave began in 2002 and contains data from 2002 to 2005, while the second wave of data covers from 2005 to 2014. Because there is no variable on the size of the firm in the first wave of data, we generate a corresponding variable for the first wave using the same classification in the latter survey as in Table A2. We also drop observations in which the firm reported "Don't know." A table of questions used in this paper can be found in Table A1. We re-coded a number of variables like WebsiteUse and OwnGenerator such that 0 implies *No* and 1

[8] The firm size categories are 5-19 employees (small), 20-99 (medium), and 100+ (large). Since the majority of firms are small and medium-sized, the surveys over-sample large firms. A full description of the data can be found at http://www.enterprisesurveys.org/methodology.

implies Yes to make interpretation of our regression coefficients more intuitive.

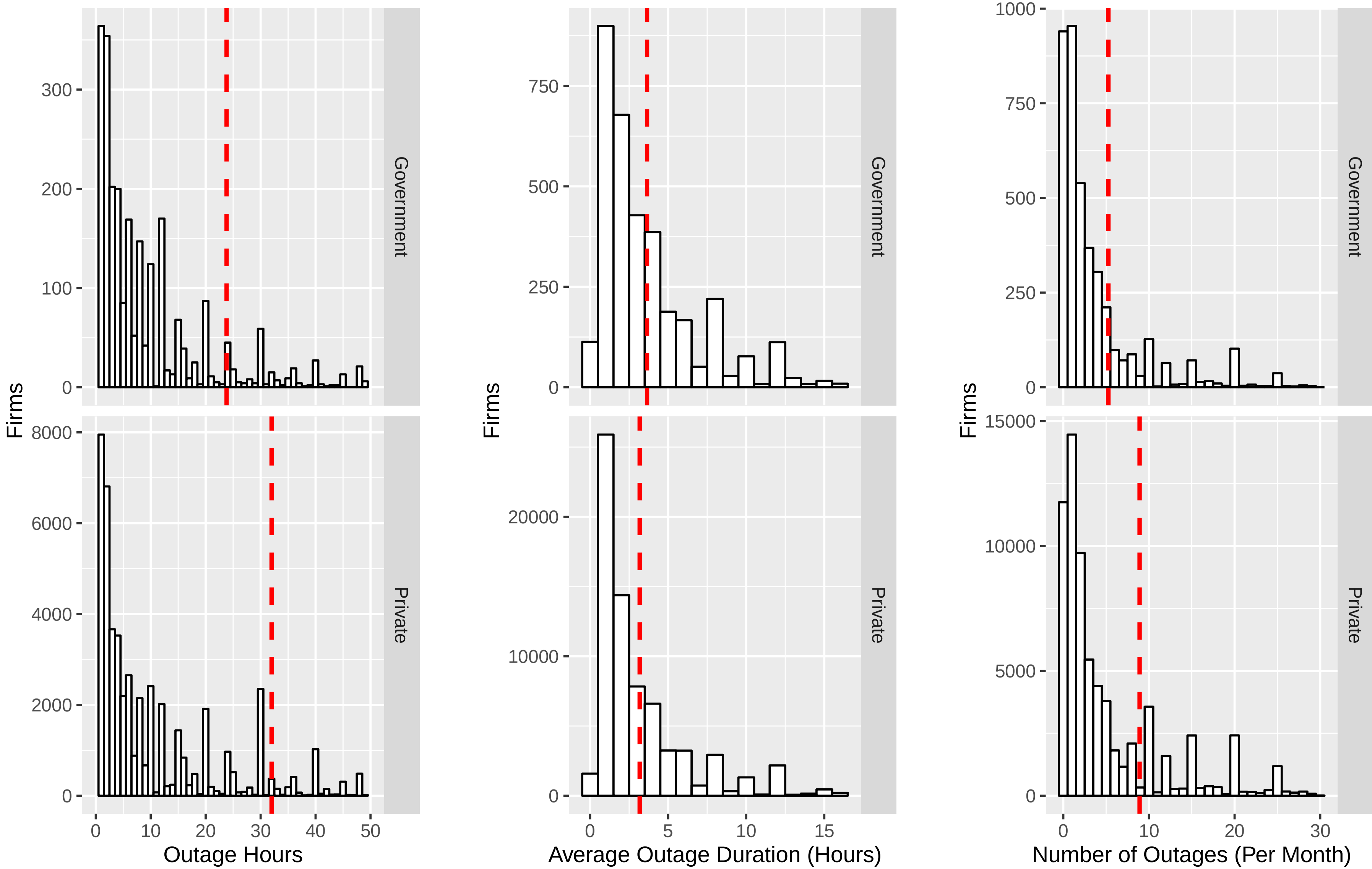

Fig. A1. Outage duration, number of outages and total outage hours by firm types.

Note: The figure plots firm-level measures of total annual outage hours, the number of outages, and their average duration by type of firm using the "energy sample."

Table A1. Questions from the World Bank Enterprise Survey

Variable	**Survey Questions/Note**	**Typical Answers**
Fin. Obstacle	Is access to financing, which includes availability and cost [interest rates, fees and collateral requirements], No Obstacle, a Minor Obstacle, a Major Obstacle, or a Very Severe Obstacle to the current operations of this establishment?	0-4; Increasing in the order of Severity
Outage No.	In a typical month, over fiscal year [insert last complete fiscal year], how many power outages did this establishment experience?	Numerical
Outage Duration	How long did the average power outage last?	Numerical
Financial Access	Transformed from FinObstacle. See Typical Answers	0-4; Increasing in the order of Easy Access
Own Generator	Whether a firm owns a generator in the previous fiscal year?	0-No 1-Yes
Website Use	Now turning to the current situation. Thinking about the present time, does this establishment use the following in its communications with clients or suppliers?	0-No 1-Yes
Transportation Obstacle	How much of an obstacle is: Tranportation of goods, supplies, and inputs?	0-4; Increasing in the order of Severity
Transportation Access	Transformed from Transportation Obstacle	0-4; Increasing in the order of Easy Access

Note: The WBES codes responses for EmailUse and OwnGenerator as 1 for yes and 2 for no, but we re-coded the answer to be consistent with the associated probit regressions.

Appendix B. Additional Tables and Figures

The following section presents additional tables and figures which characterize our data and results. Fig. A1 illustrates the different distributions of outage variables among government and private firms, which is another characteristic that we control for in the regressions. Table A2 shows the distribution of firm sizes across geographical regions in the sample. Table A1 shows the exact phrasing of the survey questions posed to firms, which make up the majority of our dependent variables, and Table A3 lists the countries included in our sample along with each country's contribution to the total number of small, medium, and large firms.

Tables A4 and A5 show results similar to Tables 2 and 6 in the main text, but with the addition of an indicator variable for whether the firm was located outside of a large city (defined as the country's capital or a city with more than a million people). Although we cannot determine the precise locations of firms (which would allow a more direct link to the location of infrastructure projects), we can differentiate between firms in large urban areas and those outside of them. Table A4 indicates that firms outside of large cities experience more outage hours and are less likely to use a website for business, as we might expect. At the same time, the interaction term between electrical investment and financial access is slightly larger when limited to firms outside of large cities, which may indicate that the importance of financial access in taking advantage of energy investments is larger for firms in smaller cities, towns, and villages. The results in A5 for the interaction term for website use are essentially unchanged when contolling for the largest cities.

Table A2. Distribution of firm sizes across regions (electricity sample vs. telecommunication sample).

Electricity sample

	Small(<20)	Medium(20-99)	Large(100 and over)	Total
East Asia Pacific	12.93	15.31	22.00	15.99
Europe and Central Asia	15.92	13.71	12.21	14.21
High income: OECD	1.41	2.10	1.82	1.76
High income: nonOECD	3.15	3.26	2.03	2.92
Latin America and Caribbean	26.29	27.66	26.15	26.76
Middle East and North Africa	0.42	0.65	0.75	0.58
South Asia	21.66	26.51	26.45	24.60
Sub Saharan Africa	18.22	10.80	8.59	13.17
Total	100.00	100.00	100.00	100.00
N	12,479	11,998	8,065	31,542

Telecommunication sample

	Small(<20)	Medium(20-99)	Large(100 and over)	Total
East Asia Pacific	9.08	11.49	15.20	11.14
Europe and Central Asia	24.79	24.54	24.66	24.68
High income: OECD	1.11	2.05	2.40	1.69
High income: nonOECD	6.01	6.21	5.06	5.88
Latin America and Caribbean	18.15	22.60	22.10	20.45
Middle East and North Africa	8.82	9.36	8.07	8.84
South Asia	6.00	6.41	9.51	6.86
Sub Saharan Africa	26.04	17.35	13.00	20.47
Total	100.00	100.00	100.00	100.00
N	52,115	39,136	24,537	115,778

Note: The table presents the distributions of firm sizes by region based on the original classifications of the World Bank. The upper panel includes firms in the "energy sample" while the lower panel includes the "telecommunications sample."

Table A3. Data Availability by Country

	Energy Sample	Telecommunication Sample
Afghanistan	0	520
Albania	168	981
Algeria	0	437
Angola	247	754
Antigua and Barbuda	0	149
Argentina	658	2064
Armenia	114	1231
Azerbaijan	0	1255
Bangladesh	865	2477
Belarus	35	1151
Belize	106	0
Benin	0	339
Bhutan	69	244
Bolivia	154	942
Bosnia and Herzegovina	0	910
Botswana	0	606
Brazil	1537	3389
Bulgaria	719	2084
Burkina Faso	0	528
Burundi	0	270
Cambodia	0	841
Cameroon	392	532
Cape Verde	0	245
Central African Republic	0	150
Chad	0	146
Chile	1104	2899
China	289	0
Colombia	413	1930
Congo	0	112
Cote d'Ivoire	219	507
Dominican Republic	0	225
Democratic Republic of Congo	0	1196
Ecuador	524	1383
Egypt	0	3582
El Salvador	479	1489
Eritrea	0	165
Ethiopia	0	639
Fiji	0	158
Macedonia	76	1047
Gabon	0	172
Gambia	140	174
Georgia	65	1063
Ghana	875	1205
Guatemala	465	1543
Guinea	0	223

Guinea Bissau	0	158
Guyana	0	325
Honduras	546	1228
India	6966	3953
Indonesia	455	2011
Iraq	0	747
Jamaica	211	365
Jordan	79	571
Kazakhstan	418	1871
Kenya	588	1685
Kosovo	0	370
Kyrgyz Republic	0	948
Laos	364	862
Lebanon	0	887
Lesotho	0	212
Liberia	55	146
Lithuania	106	685
Madagascar	204	1045
Malawi	0	306
Malaysia	351	762
Mali	0	985
Mauritania	0	236
Mauritius	0	537
Mexico	1087	2887
Micronesia	0	62
Moldova	62	1269
Mongolia	0	832
Montenegro	37	356
Morocco	106	1230
Mozambique	0	479
Nepal	0	368
Nicaragua	892	1246
Niger	0	272
Pakistan	960	1813
Panama	434	956
Paraguay	0	966
Peru	457	1626
Philippines	643	1954
Romania	677	1835
Russia	1088	6023
Rwanda	131	450
Samoa	0	107
Senegal	0	762
Serbia	0	1145
Sierra Leone	0	150
South Africa	368	1540
Sri Lanka	456	1038

Suriname	0	152
Swaziland	0	307
Syria	0	536
Tajikistan	162	1125
Tanzania	313	1442
Thailand	941	1206
Timor Leste	0	148
Togo	0	153
Tonga	0	145
Tunisia	0	592
Turkey	1344	4787
Uganda	654	1561
Ukraine	374	2748
Uruguay	0	1193
Uzbekistan	0	1329
Vanuatu	71	126
Venezuela	0	789
Vietnam	1229	2068
West Bank And Gaza	0	429
Yemen	0	753
Zambia	0	1387
Zimbabwe	0	594
N	32,542	115,788

Table A4. The effects of investments in energy infrastructure and financial access on electrical outages (with location interaction terms).

	Outage Hours		Outage Duration		Number of Outages	
	(1)	(2)	(3)	(4)	(5)	(6)
Electrical Investment (E)	-0.056***	-0.055***	-0.015**	-0.016***	-0.041***	-0.039***
	(0.010)	(0.009)	(0.006)	(0.005)	(0.008)	(0.007)
Financial Access (F)	-0.035***	-0.034***	-0.016***	-0.016***	-0.018***	-0.017***
	(0.006)	(0.006)	(0.004)	(0.003)	(0.005)	(0.004)
E x F	-0.008***		-0.003**		-0.005***	
	(0.002)		(0.001)		(0.002)	
E × F × Small City		-0.010***		-0.004***		-0.007***
		(0.002)		(0.001)		(0.002)
Small City		0.158***		-0.029		0.189***
		(0.033)		(0.021)		(0.027)
Constant	1.809***	0.890***	0.324***	1.840***	1.493***	-0.917***
	(0.043)	(0.232)	(0.026)	(0.135)	(0.035)	(0.199)
Country Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	32,542	32,542	32,542	32,542	32,542	32,542
R^2	0.590	0.591	0.282	0.282	0.685	0.685

Note: The regression estimates reported in the table are from the following specification: $ln(InfrastructureQuality_{i,t}) = \beta_1 ln(Investment_{c,t}) + \beta_2 FinancialAccess_{i,t} + \beta_3 ln(Investment_{c,t}) \times FinancialAccess_{i,t} + \beta_4 ln(Investment_{c,t}) \times FinancialAccess_{i,t} \times SmallCityDummy_i + \beta_5 SmallCityDummy_i + \varepsilon_{i,t}$. We also include dummy variables for countries and years. *FinancialAccess* is coded on a scale from 0-4 and is increasing in access. A small city is defined as a location that is not the county's capital or a city with a million people or more. The sample is restricted to firms for which full outage and national investment in energy infrastructure are available. Standard errors are in parentheses beneath the coefficient estimates. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.

Table A5. Probit estimates of the effects of investment and financial access on website use (with location interactions).

	WebsiteUse			
	(1)	(2)	(3)	(4)
Telecommunication Investment (T)	0.000	0.000	0.006	0.006
	(0.004)	(0.004)	(0.005)	(0.005)
	[0.000]	[0.000]	[0.006]	[0.006]
Financial Access (F)	0.038***	0.038***	0.022***	0.022***
	(0.003)	(0.003)	(0.003)	(0.003)
	[0.015]	[0.015]	[0.009]	[0.009]
T × F × SmallCity	0.003***	0.003***	0.004***	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)
	[0.001]	[0.001]	[0.002]	[0.002]
Small City	-0.287***	-0.287***	-0.274***	-0.274***
	(0.016)	(0.016)	(0.017)	(0.017)
Government			[-0.115]	[-0.115]
			(0.023)	(0.023)
Manufacturing			-0.259***	-0.259***
			(0.012)	(0.012)
Services			-0.048***	-0.048***
			(0.014)	(0.014)
Exporter			0.506***	0.506***
			(0.010)	(0.010)
Constant	0.197**	0.197**	-0.401***	-0.401***
	(0.093)	(0.093)	(0.098)	(0.098)
Firm Size Dummies	No	No	Yes	Yes
Country Dummies	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes
Observations	115,788	115,788	115,788	115,788
$Pseudo-R^2$	0.113	0.113	0.203	0.203

Note: Standard errors are in parentheses beneath the coefficient estimates, and marginal effects are in brackets. The probit regressions reported in the table include the full set of controls in Eq. 1 (i.e., dummy variables for firm size, ownership type, sector, and exporter status). *FinancialAccess* is coded on a scale from 0-4 and is increasing in access. A small city is defined as a location that is not the county's capital or a city with a million people or more. The sample is restricted to firms for which firm-level data on website usage and national data on investment in telecommunications infrastrcuture are available. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively.